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MH1200 Linear Algebra I

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1. The First Two Fundamental Operations on Matrices

1.1. Matrix Addition

Definition Matrix Addition — Definition 1

We define the addition of two matrices $A = \left(a_{i,j}\right)_{n\times m}$ and $B = \left(b_{i,j}\right)_{n\times m}$ with the **same shape** as:

$$A + B \coloneqq \left(a_{i,j} + b_{i,j}\right)_{n\times m}$$

Example Matrix Addition — Example 2

$$\begin{pmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{pmatrix} + \begin{pmatrix} 10 & 10 \\ 9 & 9 \\ 2 & 8 \end{pmatrix} = \begin{pmatrix} 11 & 12 \\ 12 & 13 \\ 7 & 14 \end{pmatrix}$$

1.2. Scalar multiplication

Definition Scalar Multiplication — Definition 3

We define the multiplication of a matrix $A = \left(a_{i,j}\right)_{n\times m}$ with a scalar $r \in \mathbb{R}$ as:

$$rA = Ar \coloneqq \left(ra_{i,j}\right)_{n\times m}$$

2. Properties of Matrices

Theorem Properties of Matrices — Theorem 4

1. $(M + N) + P = M + (N + P)$
2. $M + N = N + M$
3. $\mathbf{0} + M = M + \mathbf{0} = M$
4. $r(sM) = (rs)M$
5. $1M = M$
6. $0M = M$
7. $(r + s)M = rM + sM$
8. $r(M + N) = rM + rN$

Note: These properties are almost exactly the same as the conditions for a structure to be an abstract vector space. The only difference is that condition 6 is replaced by the following more general condition:

$$\forall \vec{v} \in S \exists -\vec{v} \in S \ . \ \vec{v} + (-\vec{v}) = \mathbf{0}$$

That is, that each vector has a corresponding inverse under addition.

We note that the algebra of vectors and the algebra of matrices are similar — they both follow the rules above. When mathematicians find recurring rules in distinct mathematical structures, they may abstract these rules into a more general structure (eg., abstract vector spaces). This is especially useful when the rules turn out to say quite a lot about the structure they represent — any property you prove about these general structures applies to any instance of that structure, similar to interfaces in OOP.

Example | Abstract Vector Spaces

Example 5

1. $\mathbb{R}^n$
2. $\mathbb{M}_{p\times q}(\mathbb{R})$
3. Continuous real functions
4. Sequences of numbers
5. Polynomials with degree $\leq d$
6. Linear transformations between vector spaces

Note: An **abstract vector space** equipped with a bilinear product is known as an **algebra over a field**. The idea of algebras is **very** ubiquitious in higher mathematics. Think about lie algebras in topology, clifford algebras in geometric algebra, etc...

In general, the product need not be associative. It may sometimes satisfy different properties such as the Jacobi identity though.

2.1. The Last Fundamental Operation: Matrix Multiplication

Definition | Matrix Multiplication

Definition 6

Let $C = AB$, with $A = \left(a_{i,j}\right)_{l\times n}$ and $B = \left(b_{i,j}\right)_{n\times m}$. Then,

$$AB = C \coloneqq \left(c_{i,j}\right)_{l\times m}$$

With $c_{i,j} \coloneqq \sum_{k=1}^{n} a_{i,k}b_{k,j}$

3. How Matrices Differ from $\mathbb{R}$

1. It's not always defined for two matrices
2. Matrix multiplication is non-commutative
3. Matrices don't necessarily have inverses (Given M, there may not be an M^{-1} such that $MM^{-1} = M^{-1}M = I$)
4. Matrices may have zero divisors ($AB = \mathbf{0}_{p\times q}$ even when A and B are nonzero matrices)

4. Properties of Matrix Multiplication

Theorem Existence of the Identity or Neutral Element

Theorem 7

Let:

$$I_m = \begin{pmatrix} 1 & 0 & 0 & \cdots & 0 \\ 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 1 \end{pmatrix}_{m \times m}$$

Further, let $M \in \mathbb{M}_{n\times m}(\mathbb{R})$. Then:

$$I_n M = M I_m = M$$

Theorem Associativity of Matrix Multiplication

Theorem 8

Let $A \in \mathbb{M}_{m_0\times m_1}$, $B \in \mathbb{M}_{m_1\times m_2}$, $C \in \mathbb{M}_{m_2\times m_3}$. Then:

$$A(BC) = (AB)C$$

Proof Associativity of Matrix Multiplication

Proof 9

Let $f(M, i, j)$ be the (i, j)-th entry of M. Then:

$$\begin{aligned} f(A(BC), i, j) &= \sum_{k_1=1}^{m_1} f(A, i, k_1) f(BC, k_1, j) \\ &= \sum_{k_1=1}^{m_1} f(A, i, k_1) \sum_{k_2=1}^{m_2} f(B, k_1, k_2) f(C, k_2, j) \\ &= \sum_{k_1=1}^{m_1} \sum_{k_2=1}^{m_2} f(A, i, k_1) f(B, k_1, k_2) f(C, k_2, j) \\ &= \sum_{k_2=1}^{m_2} \sum_{k_1=1}^{m_1} f(A, i, k_1) f(B, k_1, k_2) f(C, k_2, j) \\ &= \sum_{k_2=1}^{m_2} f(C, k_2, j) \sum_{k_1=1}^{m_1} f(A, i, k_1) f(B, k_1, k_2) \\ &= \sum_{k_2=1}^{m_2} f(C, k_2, j) f(AB, i, k_2) \\ &= \sum_{k_2=1}^{m_2} f(AB, i, k_2) f(C, k_2, j) \\ &= f((AB)C, i, j) \end{aligned}$$

Since all entries of $A(BC)$ and $(AB)C$ are equal,

$$A(BC) = (AB)C$$

As needed. ■

We note that one can also view matrices as linear transformations, as there is an isomorphism between these two objects. Under this view, it is more obvious why we should expect matrices to be associative — $A(BC)$ and $(AB)C$ represent the same action of applying the linear transformations corresponding to C, B, and A, in that order.

Theorem Non-commutativity of Matrix Multiplication

Theorem 10

There is some $A \in \mathbb{M}_{l\times m}$, $B \in \mathbb{M}_{m\times n}$:

$$AB \neq BA$$

In fact, in general, $AB \neq BA$.

Theorem Non-existence of Inverses

Theorem 11

There is some $A \in \mathbb{M}_{n\times n}$ such that no $B \in \mathbb{M}_{n\times n}$ exists which satisfies:

$$AB = BA = I$$

Definition Invertible Matrix

Definition 12

Consider a **square matrix** $M \in \mathbb{M}_{m\times m}(\mathbb{R})$.

We say that M is **invertible** iff there exists another matrix $M^{-1} \in \mathbb{M}_{m\times m}(\mathbb{R})$ such that:

$$MM^{-1} = M^{-1}M = I$$